

Histochemical GUS staining

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Recipe for the GUS solution (wo/ Fe and X-Gluc) (1L)

1. **1 M NaH₂PO₄**: Mix 119.98 g NaH₂PO₄ (S23or S28) and adjust volume into 1L
2. **1 M Na₂HPO₄**: Mix 141.96 g Na₂HPO₄ (S22or S24) and adjust volume into 1L
3. **0.5 M EDTA** (pH 8.0): Mix 186.12 g EDTA•2H₂O (E6) or 146.12 g EDTA (E5) and 800 mL DDW, adjust the pH to 7.8 with KOH (~20 g of KOH pellets) (need more KOH for E5), and adjust volume into 1L.

42.3 mL of 1 M NaH₂PO₄ (final conc 0.1M)

57.7 mL of 1 M Na₂HPO₄ (final conc 0.1M) → ratio for Sodium phosphate buffer pH7

20 mL of 0.5 M EDTA (final conc 0.01M)

1 mL of Triton X-100 (final 0.1%)

879 mL of DDW

Recipe for the Fe solution (50 mL) → prevent diffusion of indoxyl

1. **100 mM Potassium ferricyanide**: Mix 1.645 g Potassium ferricyanide (P20 and P21) and adjust volume to 50 mL DW. (0.33g / 10ml)
2. **100 mM Potassium ferrocyanide (Potassium hexacyanidoferrate(II) trihydrate)**: Mix 2.112 g Potassium ferrocyanide (P25) and adjust volume to 50 mL DW. (0.42g / 10ml)

*always fresh

Recipe for the GUS solution (50 mL, prepare before use)

1.5mL of **100 mM Potassium ferrocyanide** (final conc 1.5 mM, recommend 0.5-5 Mm)

1.5mL of **100 mM Potassium ferricyanide** (final conc 1.5 mM, recommend 0.5-5 mM)

0.5 mL of **100 mM X-Gluc** (in DMSO)

46.5 mL of **the GUS solution** (wo/ Fe and X-Gluc)

GUS observation

1. Incubate seedling in GUS solution for in 37 °C
2. Rinse in 70% ethanol for overnight
3. Rinse in 100% ethanol for overnight
4. Observe in 100% ethanol

(if necessary)

Remove chlorophyll in Ethanol:Acetic acid = 6:1 several times

Observe in Viskol for transparency